

## Nine times table number grid

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 |
| 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 |
| 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 |
| 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 |
| 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 |
| 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 |
| 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 |
| 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 |

Look what happens if you put the answers to the  $9\times$  table in a grid.



The answers form a diagonal line!



## The nine times table made easy

This is a rule that works for all the answers in the nine times table between  $2 \times 9$  and  $10 \times 9$ . The key is to do some quick mental subtraction.

### STEP 1

The answers start with a digit one less than the number you are multiplying by 9.

### STEP 2

The second digit of the answer is equal to 9 minus the first digit.

$$2 - 1 = 1$$

$$2 \times 9 = 18$$

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$$9 - 1 = 8$$

$$3 - 1 = 2$$

$$3 \times 9 = 27$$

$$3 \times 9 = 27$$

$$9 - 2 = 7$$



## Cats and kittens

If 9 cats each have 9 kittens, how many cats will there be in all?

## TOP TIP

Here's an easy way of finding out whether a number is a multiple of nine. You can use this to check your answers. If a number is a multiple of nine, then the digits in the answer will add up to nine.

$$36 \rightarrow 3 + 6 = 9$$

This works with bigger numbers, too. If you add together the numbers in the answer and get a two-digit number, then you need to add those digits together too.

$$99 \rightarrow 9 + 9 = 18$$

$$1 + 8 = 9$$

Can you work out whether **783** and **16,947** are multiples of nine?